System
System.IO: https://hackage.haskell.org/package/base/docs/System-IO.html

System.IO

```
-- System.IO
data IO a
type FilePath = String

--
readFile :: FilePath -> IO String
readFile' :: FilePath -> IO String
writeFile :: FilePath -> String -> IO ()
appendFile :: FilePath -> String -> IO ()
putChar :: Char -> IO ()
putStr :: String -> IO ()
getChar :: IO Char
getLine :: IO String
getContents :: IO String
interact :: (String -> String) -> IO ()


-- handle
data Handle
stdin :: Handle
stdout :: Handle
stderr :: Handle
data IOMode = ReadMode | WriteMode | AppendMode |
ReadWriteMode

openFile :: FilePath -> IOMode -> IO Handle
withFile :: FilePath -> IOMode -> (Handle -> IO a) -> IO
a
hClose :: Handle -> IO ()
hIsEOF :: Handle -> IO Bool
hFileSize :: Handle -> IO Int
hSetFileSize :: Handle -> Int -> IO ()
hTell :: Handle -> IO ()
hIsOpen :: Handle -> IO Bool
hIsClosed :: Handle -> IO Bool
hIsReadable :: Handle -> IO Bool
hIsWritable :: Handle -> IO Bool
hIsSeekable :: Handle -> IO Bool
hIsReady :: Handle -> IO Bool
hPutChar :: Handle -> Char -> IO ()
hPutStr :: Handle -> String -> IO ()
hPutStrLn :: Handle -> String -> IO ()
hPrint :: Show a => Handle -> a -> IO ()


-- Buffering
data BufferMode = NoBuffering | LineBuffering |
BlockBuffering (Maybe Int)
hSetBufferMode :: Handle -> BufferMode -> IO ()
hGetBufferMode :: Handle -> IO BUfferMode
```

```
hFlush :: Handle -> IO ()


-- Positioning
data HandlePosn
hGetHandlePosn :: Handle -> IO HandlePosn
hSetHandlePosn :: HandlePosn -> IO ()

data SeekMode = AbsoluteSeek | RelativeSeek |
SeekFromEnd
hSeek :: Handle -> SeekMode -> Integer -> IO ()


-- Temp file
openTempFile :: FilePath -> String -> IO (FilePath,
Handle)
openBinaryTempFile :: FilePath -> String -> IO
(FilePath, Handle)
openempFileWithDefaultPermissions :: FilePath -> String
-> IO (FilePath, Handle)
openBinaryTempFileWithDefaultPermissions :: FilePath ->
String -> IO (FilePath, Handle)


-- Encoding
data TextEncoding
latin1 :: TextEncoding
utf8 :: TextEncoding
utf16 :: TextEncoding
utf32 :: TextEncoding
...
hGetTextEncoding :: Handle -> IO (Maybe TextEncoding)
hSetTextEncoding :: Handle -> TextEncoding -> IO ()
mkTextEncoding :: String -> IO TextEncoding


-- Newline
data Newline = LF | CRLF
data NewLineMode = NewLineMode { inputNL :: NewLine,
outputNL :: NewLine }
universalNewLineMode :: NewLineMode
nativeNewLineMode :: NewLineMode
noNewLineTranslation :: NewLineMode
hSetNewLineMode :: Handle -> NewLineMode -> IO ()
```

System.Env

```
getArgs :: IO [String]
getProgName :: IO String
getExecutablePath :: IO FilePath
withArgs :: [String] -> IO a -> IO a
withProgName :: String -> IO a -> IO a
```

```
getEnv :: String -> IO String
setEnv :: String -> String -> IO ()
unsetEnv :: String -> IO ()
lookupEnv :: String -> IO (Maybe String)
getEnvironment :: IO [(String, String)]
```